Smoothness-indexed penalized trend estimation on measured vehicle trajectories: jerk, conditional extrapolation and selection-method comparison

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ABSTRACT

We apply finite-difference penalized least squares to real, noisy vehicle-position data for which third differences have an interpretable connection to jerk. Five camera-derived longitudinal vehicle segments from the Federal Highway Administration NGSIM US-101 dataset are selected deterministically and sampled at 10 Hz. A normalized smoothness coordinate $S \in [0, 1]$ controls polynomial trend regularization for orders $d = 1, 2, 3, 4$. Five classical selectors and two mean-difference policies are compared through strictly chronological observations held out for the next 0.5, 1, and 2 seconds; constant-velocity, constant-acceleration and Savitzky–Golay extrapolations provide physical baselines. We explicitly append a local polynomial continuation to the fitted PLS trend: forecasting is not an intrinsic output of PLS. An additional matched experiment compares a third-difference three-point forecast recurrence with the original 25-point continuation and GCV-selected natural cubic smoothing splines; their position-forecast rankings vary by horizon. Seven numerical implementations are compared separately for fitting cost and agreement.Seven numerical implementations are compared separately for fitting cost and agreement. No future information enters selection, no synthetic data or noise are injected, and no ground-truth jerk measurement is claimed. This is an illustrative and falsifiable physical-data case study rather than a vehicle-control guarantee.

1. Introduction

Vehicle trajectories provide a physical interpretation for higher-order trend smoothing: the first three time derivatives of longitudinal position are velocity, acceleration and jerk. The US Federal Highway Administration's Next Generation Simulation (NGSIM) dataset reports camera-derived vehicle positions every 0.1 seconds (U.S. Department of Transportation, Federal Highway Administration, 2016). Earlier vehicle-trajectory research emphasizes that estimating acceleration from these observations magnifies measurement noise (Thiemann, Treiber and Kesting, 2008); third derivatives amplify it further. This motivates examining a third-difference penalty ($d = 3$), rather than restricting the study to the commonly used second-difference penalty ($d = 2$).

We study the PLS model of Guerrero (2007), its trace-based smoothness index (Guerrero, 2008) and classical tuning criteria (Cortés-Toto, Guerrero and Reyes, 2017). Companion papers examine numerical accuracy and computational complexity. Here we focus on a limited empirical study using recorded vehicle trajectories. We examine the estimated historical trends and assess how a *separate* local extrapolation rule predicts held-out vehicle positions.

We also compare two boundary continuation rules for the same fitted $d = 3$ PLS trend and benchmark them against natural cubic smoothing splines with GCV-selected penalties, Savitzky–Golay differentiation and local polynomial fits. These are standard local/global families for estimating derivatives from noisy experimental data (Ahnert and Abel, 2007), rather than independent numerical matrix-factorization methods. The resulting analysis tests an implementation and application of an established PLS model, not an assertion of historically first use of $d = 3$ differences.

The dataset supplies physical position, but not independently measured latent truth or jerk.The dataset supplies physical position, but not independently measured latent truth or jerk. We do not treat estimated jerk as a direct measurement, add synthetic noise or conflate smoothing, parameter selection, extrapolation and held-out evaluation.

1.1. The noise amplification motivating third-order smoothing

If measurement noise ε_t has independent variance σ_ε^2, third forward differentiation uses stencil $(-1, 3, -3, 1)$, hence

$$\mathrm{Var}\left(\frac{D_3\varepsilon_t}{(\Delta t)^3}\right) = \frac{20\sigma_\varepsilon^2}{(\Delta t)^6}. \tag{1}$$

With $\Delta t = 0.1$ s, direct third differences can be extremely unstable. Equation (1) is a conditional mathematical illustration, not an empirical noise-variance estimate: camera-tracking errors may be temporally correlated. Regularization suppresses variation but can bias sudden real changes in jerk. Thus plotted derivatives remain *estimates*, not ground truth.

2. Observed data and reproducible design

2.1. Camera-derived NGSIM US-101 longitudinal tracks

The source is the US DOT/Federal Highway Administration NGSIM vehicle trajectory dataset (DOI: 10.21949/1504477)

ORCID(s):

(U.S. Department of Transportation, Federal Highway Administration, 2016). Its `Local_Y` field is the longitudinal front-center vehicle coordinate, recorded in feet; `Global_Time` is in milliseconds. We convert feet to meters exactly by 0.3048 and use elapsed time since each segment begins. The official dataset indicates CC BY-SA 3.0; extracted position subsets carry attribution and immutable hashes.

Data are downloaded directly from the USDOT Socrata API using a fixed query for US-101 vehicle identifiers 1–20. After sorting by vehicle and timestamp, a 100-ms gap splits track segments. In deterministic ascending order, the first five segments meeting minimum duration (200 points), displacement (20 m), and conservative gross-quality thresholds are kept, without looking at forecast accuracy. The original vehicle identifiers may occur in several distinct recording periods; we retain track start timestamps to distinguish them. The complete extraction rule and source SHA-256 digests reside with the research scripts. Raw positions, rather than an already smoothed NGSIM derivative series, are the input.

The windows contain $n = 80$ consecutive observations (8 s), with prediction origins at segment indices 95, 140 and 185. Futures of $h = 5, 10, 20$ frames (0.5, 1 and 2 s) are withheld chronologically. The 15 origin windows overlap within five recorded segments. Observed held-out position is the error target, *not* unavailable latent noise-free position or actual jerk; aggregate error scores are descriptive and not independent-sample confidence intervals.

2.2. Smoothness-indexed quadratic trend

For forward differences D_d with $d \in \{1, 2, 3, 4\}$, and fixed numeric mean of differences μ, consider

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y - \tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}. \tag{2}$$

If $\delta_j > 0$ are the $n - d$ positive eigenvalues of $D_d^\top D_d$, the normalized degree-of-smoothing index is

$$S_d(\lambda) = \frac{1}{n-d} \sum_{j=1}^{n-d} \frac{\lambda\delta_j}{1 + \lambda\delta_j} \in [0, 1]. \tag{3}$$

It is continuous and strictly increasing for $\lambda > 0$, and independent of observed y and fixed μ. We invert S once on the shared 16-point grid, including exact $S = 0$ and $S = 1$ endpoints, then reuse penalties across methods. No independently spaced log-λ grid is a competitor here.

For $d = 3$, the finite difference $D_3\tau_t$ divided by $(\Delta t)^3$ approximates physical jerk of a position signal. At $S = 1$, fixed $\mu = 0$ produces a quadratic trend (constant acceleration), whereas a fixed $\mu \neq 0$ produces a cubic trend with approximately constant nonzero jerk. At finite S, the estimated third differences can vary over time. The two investigated policies are fixed $\mu = 0$ and training-only plug-in $\widehat{\mu} = \text{mean}(D_d y_{\text{train}})$. Conditional CV scores for the latter do not re-estimate μ after each observation deletion, so cannot be interpreted as exactly refitted LOOCV.

2.3. Five smoothing criteria

On each training window and difference order, we minimize ordinary fixed-μ observation-deletion CV, generalized CV (GCV), AICc, residual-based BIC and Gaussian profiled marginal likelihood (ML) over the same bounded S candidates. Residual BIC is explicitly restricted to $S \geq 0.10$ because its unrestricted interpolation boundary is ill-posed; this is a sensitivity choice, not a theoretical correction. The Gaussian criterion profiles observation variance in a variance-components model with $\lambda = \sigma_\varepsilon^2/\sigma_\eta^2$, treating the polynomial nullspace as a nuisance component. A plug-in $\widehat{\mu}$ is not joint ML estimation of all variance and mean parameters. As a result, the criterion rows describe *distinct in-sample selectors*, not directly comparable likelihood functions.

2.4. Extrapolating a trend is a separate forecasting heuristic

The public PLS fitting API estimates only the observed-period trend. For every selected fit, we use the final 25 smoothed points and fit a local polynomial of degree $d - 1$ after removing the particular solution $\mu\binom{t}{d}$, then extrapolate that polynomial and add the particular solution back at future indices. This imposes the local limiting-d polynomial pattern but *is not* a model-based predictive distribution or an intrinsic TrendEstimation forecasting algorithm. It can fail when the next driver action changes acceleration.

All methods receive identical held-out future positions. Physical comparators are 20-point local linear (constant-velocity) extrapolation, 30-point local quadratic (constant-acceleration) extrapolation, and a 21-point order-three Savitzky–Golay smoother followed by a 20-point local linear continuation (Savitzky and Golay, 1964). Forecast RMSE is the square root of mean squared position errors in meters; MAE is likewise in meters.

2.5. Three-point recurrence and spline-based forecast comparators

Following the published trend-extrapolation construction for lower orders in Guerrero (2007), we investigate its direct algebraic extension to $d = 3$. Write the final fitted trend as $\widehat{\tau}_N$ and its backward differences as $\nabla\widehat{\tau}_N = \widehat{\tau}_N - \widehat{\tau}_{N-1}$ and $\nabla^2\widehat{\tau}_N = \widehat{\tau}_N - 2\widehat{\tau}_{N-1} + \widehat{\tau}_{N-2}$. Imposing a constant third difference equal to the fitted reference mean μ beyond the observation boundary yields

$$\widehat{\tau}_N(h) = \widehat{\tau}_N + h\nabla\widehat{\tau}_N + \binom{h+1}{2}\nabla^2\widehat{\tau}_N + \binom{h+2}{3}\mu, \qquad h \geq 1. \tag{4}$$

This $d = 3$ recurrence is derived here as a forecasting rule from the constant-difference condition, rather than claimed as a new PLS estimator or a formula explicitly printed by Guerrero. It uses only the last three *fitted* historical values; the previously described local-polynomial continuation uses the last 25. For each trial the fitted PLS trend, λ, S and μ are identical across the two continuation rules.

For comparison with established local and global numerical differentiation methods (Ahnert and Abel, 2007; Cao, Bryant, Molteno, Fox and Parry, 2021), we fit a *natural cubic smoothing spline* to the same training positions, selecting its second-derivative penalty by training-only generalized cross-validation (SciPy's `make_smoothing_spline`). Four predeclared continuations use second- or third-order Taylor terms from its endpoint derivatives, a local quadratic fitted to the last 25 spline-smoothed positions, or the spline's native polynomial extrapolation. Secondary comparisons include a separately specified cubic spline with fixed training-only noise-scale heuristic, Savitzky–Golay endpoint derivatives, and local quadratic or cubic polynomial fits. No spline tuning parameter is selected from future observations. Each approach forecasts *position*, not independently measured jerk.

2.6. Separately comparing numerical implementations

On the same recorded training windows at $d = 3$, $S = 0.90$, with both μ policies, we evaluate seven pathways: full shift-free SVD, fresh dense solve, dense Cholesky, general banded LU, banded Cholesky, symmetric Gram eigendecomposition and dual SVD. Each has three within-run wall-clock repetitions and a numerical fit discrepancy measured against full SVD. These are *one-fit* timings; they do not include the computational cost of evaluating full CV/GCV/AICc/BIC/ML grids. Any failed or nonfinite pathway is reported as unavailable rather than treated as a fast winner.

3. Real-data results and comparisons

3.1. One held-out trajectory and its fitted jerk

Figure 1 shows a predeclared chronological cutoff of the first real US-101 vehicle segment. The original recorded positions, smooth in-sample trend and extrapolated future are shown separately. Predictions are made without seeing the future; the PLS fit itself contains no built-in forecast model. Third differences have meaningful units only after dividing by the cubed 0.1-s sampling interval. Figure 2 illustrates strong differentiation noise in the original position and the smaller, regularized jerk proxy of the selected PLS trend. Neither derivative is independently observed true jerk.

3.2. Five criteria, four orders and both mean policies

Table 1 reports genuinely held-out position RMSE at the one-second horizon for the main PLS orders and physical baselines; Figure 3 provides the complete $d = 1, \ldots, 4$ by five-criterion comparison for both $\mu = 0$ and training-data plug-in mean of differences. Every method is scored against the same recorded future positions, not an unavailable noise-free latent trend. For $d = 3$, GCV-selected PLS continued from 80 observed frames gives 0.583 m RMSE under fixed $\mu = 0$ and 0.475 m when the training-only sample mean of third differences is plugged in. The corresponding constant-velocity and constant-acceleration baselines give 0.547 m and 0.626 m, respectively. The comparison is illustrative; highly similar criterion-level scores do not indicate five independent demonstrations of superiority.

3.3. Extrapolation horizon and physical baselines

The experiment evaluates $h = 5, 10, 20$ future 100-ms frames rather than labeling in-sample fitted values as forecasts. Figure 4 compares selected $d = 2, d = 3$ PLS continuations with constant-velocity, constant-acceleration and Savitzky–Golay physical baselines. Smooth historical fits can extrapolate poorly if acceleration changes abruptly.

3.4. Three-point $d = 3$ continuation versus smoothing splines

A focused, matched follow-up compares the third-difference recurrence in Eq. (4) with the original 25-point continuation using exactly the same archived $d = 3$ trend fits and in-sample GCV selections. Conventional natural cubic smoothing splines independently select their penalty by GCV on the 80 training positions. The new numerical results were computed on the named self-hosted Windows runner; original experimental records were left intact.

The three-point recurrence performs best at the 0.5-s horizon and nearly matches the GCV spline's quadratic continuation at 1 s (0.2958 versus 0.2951 m). At 2 s the latter has lower observed-future RMSE (0.8085 versus 1.2998 m). The relative merit of a nonzero third-difference reference depends on the forecast rule: it improves the local 25-point continuation on these observations but does not improve the three-point recurrence. These comparisons are historical, exploratory results for only two vehicle identifiers, not independent validation of a population-level advantage or of the accuracy of estimated physical jerk. Raw outcome rows, environment manifests and replication scripts are archived under `experiments/experiment001`, `experiment002`, and `experiment003`.

3.5. All seven numerical fitting pathways

For completeness, we separately compare shift-free full SVD, dense solve, dense Cholesky, general banded LU, banded Cholesky, symmetric Gram eigendecomposition and dual SVD using the same observed windows at $d = 3, S = 0.9$. Table 3 and Figure 5 give fit-only median timings and relative fit discrepancies against SVD. This is a numerical calculation study, not a five-criterion selection runtime study or a measurement of physical jerk accuracy.

4. Discussion and conclusions

The vehicle-position data show why difference order, reference mean and extrapolation rule must be distinguished. For a position series, $d = 3$ has a direct relation to the third temporal derivative—jerk—but neither the raw third difference nor its smoothed counterpart is an independent physical reference signal. A smooth estimated jerk curve may miss abrupt maneuvers.

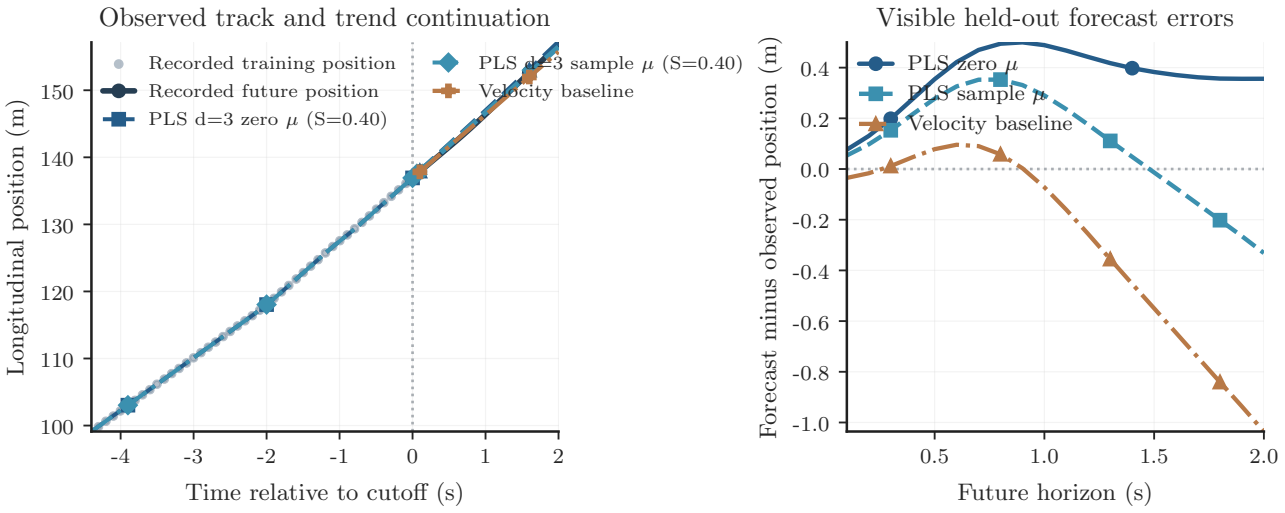


Figure 1: Measured training and future longitudinal position, with $d = 3$ GCV-selected PLS trends under zero and training-derived means and their local-polynomial continuations. The observed future is withheld during estimation.

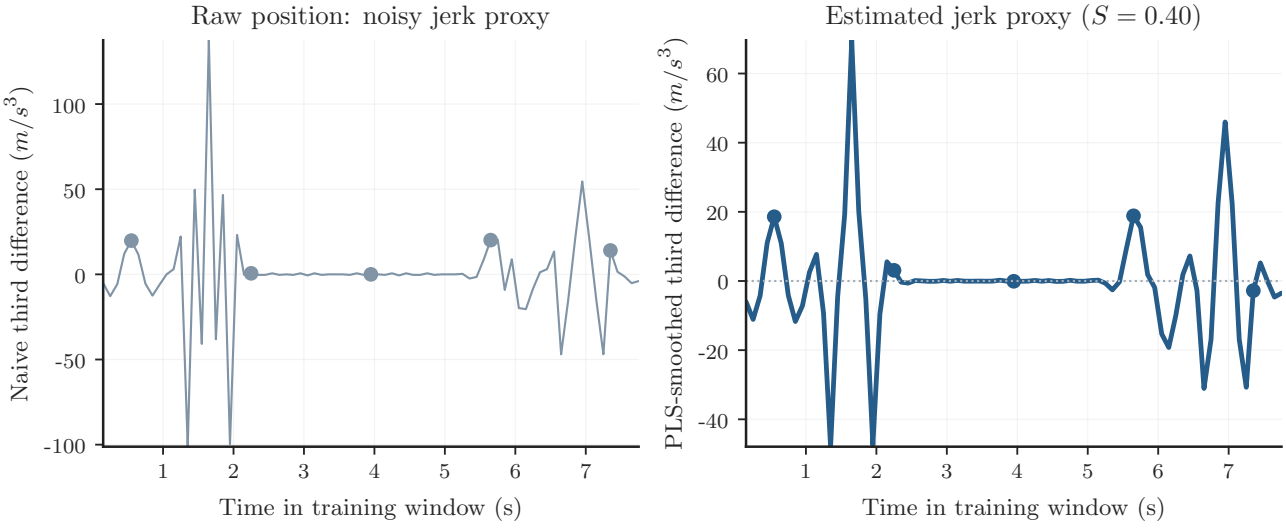


Figure 2: Direct and PLS-smoothed third differences of the same measured US-101 position segment. Panel axes have different scales; reduced amplitude after smoothing does not establish physical accuracy.

The normalized coordinate $S \in [0, 1]$ represents the fraction of penalizable effective degrees of freedom removed. Its definition is consistent across difference orders, and its inverse is independent of a fixed reference-difference mean. The user-facing selection grid is always expressed in S, not an arbitrary independently sampled log-λ coordinate. Pretranslation and reuse are computational optimizations, not new statistical models.

The favorable one-second held-out result for GCV-selected $d = 3$ is descriptive. This small sample does not establish a general forecasting advantage for vehicle trajectories. The fifteen training/holdout origins overlap within five recorded segments and do not justify naive independent-window inference. Selection criteria may yield nearly identical smooth trends, and the observed position itself includes unmeasured camera noise.

The training-data mean-difference policy uses no future information, but CV scores computed conditionally on that plug-in mean are not exact re-estimated LOOCV. Profiled Gaussian marginal likelihood requires variance assumptions; it is not point-estimator MLE for τ or joint ML estimation of μ. Restricted residual BIC remains an exploratory constrained criterion because the original unrestricted interpolation boundary is mathematically pathological.

The matched recurrence and GCV-spline comparisons sharpen this limitation. At the shortest horizon, the $d = 3$ PLS recurrence using $\mu = 0$ yields the lowest observed-future position RMSE among the reported methods; at two

Table 1
Out-of-sample position RMSE in meters at a one-second horizon (15 chronological track-origin windows). The five selection criteria used the same normalized S candidates, except restricted BIC ($S \geq 0.10$). The data-derived mean policy changes model specification and does not yield exact refitted LOOCV.

Selector	$d = 2,\ \mu = 0$	$d = 3,\ \mu = 0$	$d = 3,\ \widehat{\mu}$	$d = 4,\ \mu = 0$
CV	0.600	0.583	0.475	0.933
GCV	0.600	0.583	0.475	0.933
AICc	0.600	0.583	0.475	0.932
Restricted BIC	0.600	0.583	0.475	0.933
Profile ML	0.600	0.583	0.475	0.932
Velocity baseline		0.547		
Acceleration baseline		0.626		
Savitzky–Golay velocity		0.543		

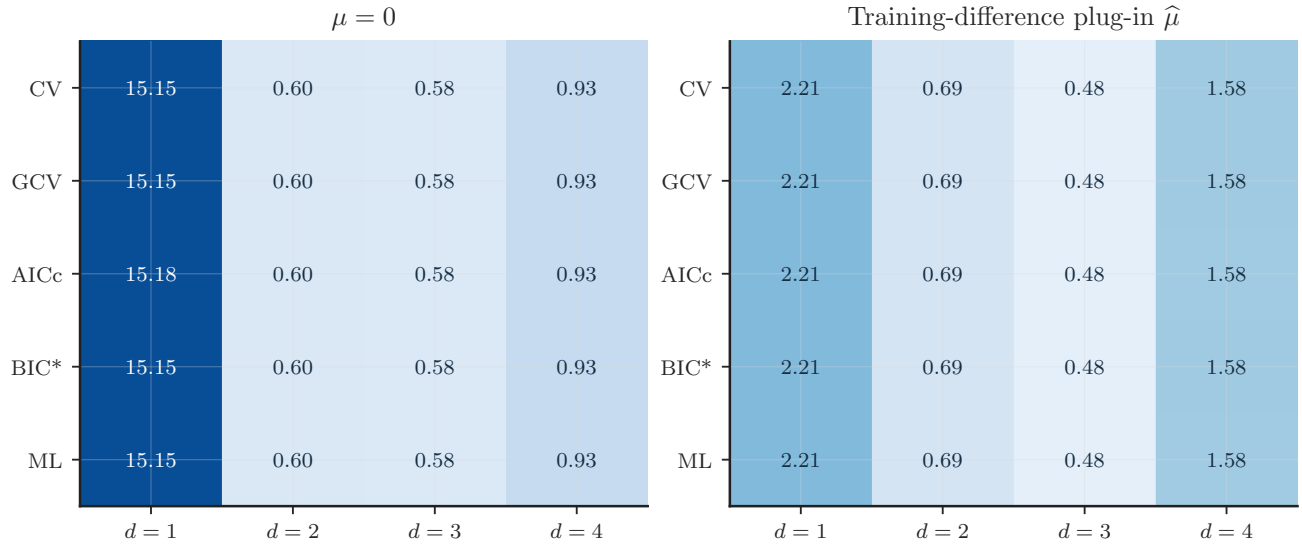


Figure 3: Held-out one-second position RMSE in meters (printed values) across all four orders and all five criteria; μ policies are separated. Colors represent log-RMSE for display, not significance. BIC was restricted to $S \geq 0.10$.

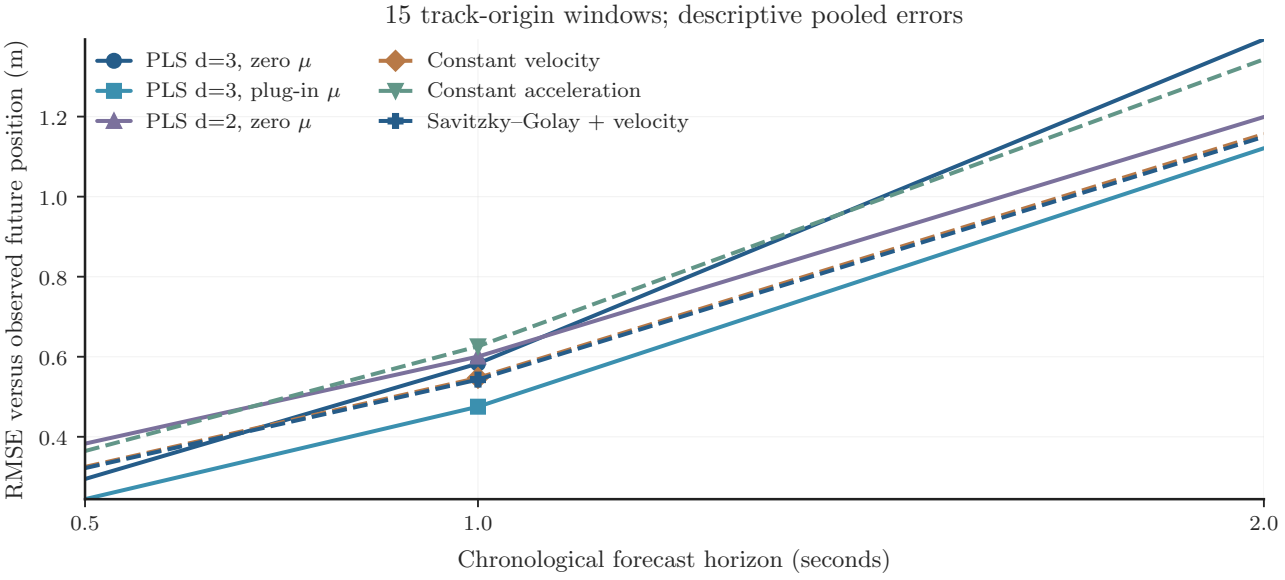


Figure 4: Chronological 0.5, 1 and 2-second position errors across 15 overlapping track-origin windows; comparisons use future measured camera-derived position, not latent truth.

Table 2
Descriptive observed-future position RMSE (meters) at three horizons for the $d = 3$ PLS forecast variants and established alternatives. All methods use the same 15 chronological origins across five track segments. The segments correspond to only two distinct vehicle identifiers; overlapping origins prevent independent-sample significance claims. Spline GCV parameters are selected on the training windows only. Smaller values are preferable.

Method and reference policy	0.5 s	1.0 s	2.0 s
PLS, three-point recurrence, $\mu = 0$	**0.0745**	0.2958	1.2998
PLS, three-point recurrence, training $\widehat{\mu}$	0.0769	0.3113	1.4006
PLS, local 25-point continuation, $\mu = 0$	0.2945	0.5831	1.3928
PLS, local 25-point continuation, training $\widehat{\mu}$	0.2441	0.4752	1.1210
Natural cubic smoothing spline (GCV), quadratic Taylor	0.1233	**0.2951**	**0.8085**
Natural cubic smoothing spline (GCV), local 25-point	0.2945	0.5830	1.3926
Savitzky–Golay, quadratic Taylor	0.2840	0.5704	1.3072

Table 3
Same observed position windows, seven fit-only numerical backends, $d = 3$, $S = 0.90$. Median runtime is across 10 model/mean cases, each with three within-run timing repetitions; maximum relative fit error uses full SVD as the numerical baseline. Hardware-specific values do not predict criterion-selection timing.

Solver	Valid cases	Median fit time (ms)	Max. relative discrepancy
Full SVD	10	0.0423	0.00e+00
Dense LU	10	0.0566	1.38e-12
Dense Cholesky	10	0.0401	1.65e-12
Banded LU	10	0.2047	1.27e-12
Banded Cholesky	10	0.0192	1.07e-12
Gram eigendecomp.	10	0.4226	3.26e-12
Dual SVD	10	0.6657	2.62e-17

seconds the GCV-selected spline with a quadratic continuation performs better. Moreover, the training-derived nonzero reference mean helps the 25-point continuation in the present pilot but not the three-point recurrence. Thus an apparent benefit of $\mu \neq 0$ cannot be attributed solely to smoothing without separating the historical estimator from the extrapolator. The five track segments correspond to only *two distinct vehicle identities*; replication on independent vehicles is essential before generalizing these numerical rankings.

Forecasts come from a local polynomial fitted to the historical PLS trend.Forecasts come from a local polynomial fitted to the historical PLS trend. It is *not* a predictive distribution, and it cannot anticipate unobserved changes in

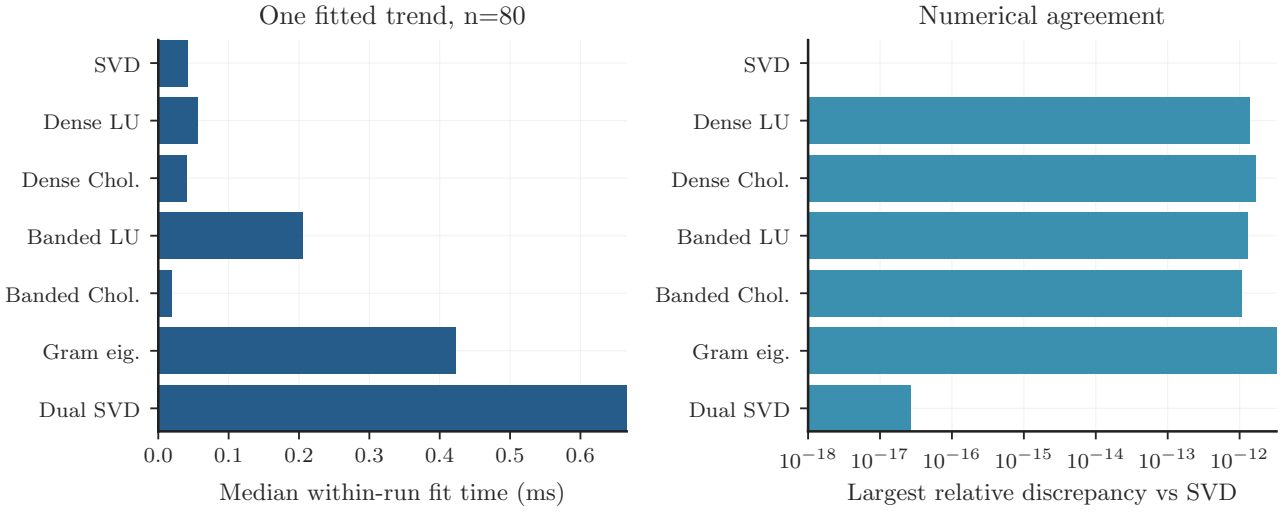


Figure 5: Fit-only elapsed milliseconds and relative numerical discrepancies on real recorded positions under two μ policies. Numerical fit differences are not statistical forecast errors.

acceleration, traffic interactions or lane changes. Future research requires independent maneuvers, substantially more vehicles, genuine measurement-error characterization, and a dedicated dynamic/state-space forecast comparison. The seven fitting backends are benchmarked for single-curve cost and numerical agreement; complete-selection runtime and independent multiprecision accuracy are addressed by companion studies.

Reproducibility and data rights

The US DOT/FHWA NGSIM dataset (DOI: 10.21949/1504477) is credited as the sole source of observed positions; the dataset portal identifies a CC BY-SA 3.0 license. Source query and extraction rules, tracked vehicle identifiers, segment starting timestamps, immutable checksums, chronological forecast rows, solver diagnostics, source-generation scripts and editable PGF vector figures are preserved under the case-study directory. The original official-position subset has not been artificially contaminated or pre-smoothed. Figure-derived jerk is explicitly estimated rather than observed.

Declarations

This is an illustrative research draft. Authorship, correspondence, funding, conflicts and any journal submission require coauthor review and agreement.

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